

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA5112

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

Industry Problem-Solving Task

Code of Conduct:

I G. Srivalli (192365014) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G. Srivalli

Annexure A

1. SSL/TLS Overhead and Web Server Performance Optimization

A web server uses TLS 1.3 for secure communication. Each HTTP response carries 2000 bytes of payload. TLS adds 25 bytes (record header + authentication tag) and handshake overhead of 3 KB per session. The server handles 10,000 requests per second, with 1,000 new TLS handshakes per second.

Task:

- A. Calculate the total data transmitted per second including TLS overhead.
 - B. Determine the percentage overhead due to TLS encryption.
 - C. Estimate bandwidth consumed due to handshake overhead per second.
 - D. Analyze the trade-off between security and latency, and propose optimization techniques (e.g., session resumption, HTTP/2, or QUIC).
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2. PGP-Based Email Encryption Load in Enterprise System

An enterprise email server encrypts emails using PGP. Each email contains a 10 MB attachment. PGP introduces 800 bytes overhead for key encryption and signatures. The system processes 5,000 emails per day.

Task:

- A. Calculate the total data transmitted per day including overhead.
 - B. Determine the percentage increase in storage due to PGP.
 - C. Analyze the computational overhead during encryption/decryption.
 - D. Recommend optimization strategies (e.g., compression, batching, or hardware acceleration).
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3. IPSec Transport vs Tunnel Mode Performance Comparison

A company compares IPSec transport mode and tunnel mode for securing internal communications. Original packet size = 1000 bytes payload + 20-byte header. Transport mode adds 30 bytes, while tunnel mode adds 50 bytes overhead. Traffic volume = 20,000 packets per second.

Task:

- A. Compute total packet size for both modes.
 - B. Calculate bandwidth usage for each mode per second.
 - C. Compare efficiency and security benefits of both modes.
 - D. Recommend the appropriate mode for internal vs external communication with justification.
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4. S/MIME Certificate Management in Large Organization

An organization manages 12,000 employees using S/MIME certificates. Each certificate is 2 KB, and certificate renewal happens every year. During renewal, both old and new certificates are stored simultaneously for 1 month.

Task:

- A. Calculate total storage required during the renewal period.
 - B. Estimate certificate distribution overhead across the network.
 - C. Analyze challenges in certificate lifecycle management at scale.
 - D. Propose optimization strategies (e.g., automation, centralized PKI, or short-lived certificates).
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5. Email Encryption Latency in Cloud-Based Systems

A cloud email service encrypts outgoing emails using AES (data) and RSA (key exchange). AES encryption takes 0.2 ms per MB, and RSA key exchange takes 5 ms per email. Each email has 8 MB attachment. The system processes 8,000 emails per hour.

Task:

- A. Calculate total encryption time per email.
- B. Estimate total processing time per hour.
- C. Analyze the performance bottleneck between symmetric and asymmetric operations.
- D. Recommend optimization strategies (e.g., ECC adoption, parallel processing, or hardware acceleration).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

QUESTION 1: SSL/TLS Overhead and Web Server Performance Optimization

Problem Statement:

A web server uses TLS 1.3 for secure communication. Each HTTP response carries 2,000 bytes of payload. TLS adds 25 bytes (record header + authentication tag) and handshake overhead of 3 KB per session. The server handles 10,000 requests per second, with 1,000 new TLS handshakes per second.

Tasks: A. Calculate total data transmitted/sec. B. Determine % overhead due to TLS encryption. C. Estimate bandwidth consumed due to handshake overhead/sec. D. Analyze security vs latency trade-off & propose optimization techniques.

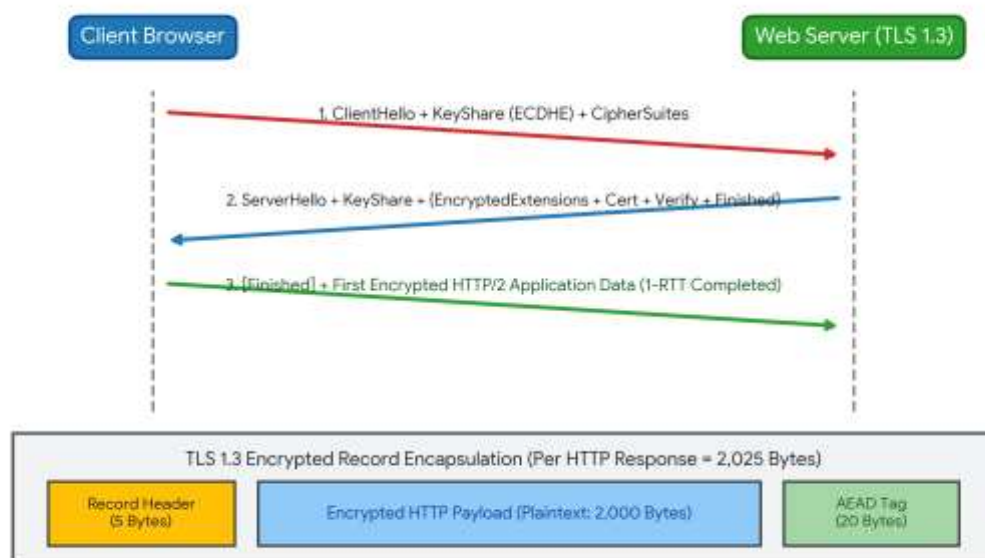


Figure 1.1: TLS 1.3 1-RTT Handshake Flow and Record Layer Encapsulation Structure

Part A: Calculation of Total Data Transmitted Per Second

To determine the total data transmitted per second, we account for two distinct traffic components: the Application Data Transmission (carrying the HTTP payload encapsulated in TLS records) and the Session Handshake Traffic (establishing cryptographic security associations).

1. Application Payload Data Volume:

- Request volume = 10,000 requests/sec
- HTTP payload per response = 2,000 bytes
- Payload Data = 10,000 requests/sec × 2,000 bytes = 20,000,000 bytes/sec = 20.00 MB/sec (160.00 Mbps)

2. TLS Record Encryption Overhead Volume:

- TLS record header + AEAD authentication tag = 25 bytes per record
- Record Overhead Data = 10,000 requests/sec × 25 bytes = 250,000 bytes/sec = 0.25 MB/sec (2.00 Mbps)

- Total Application Data Layer = 20,000,000 + 250,000 = 20,250,000 bytes/sec (20.25 MB/sec)

3. TLS Handshake Traffic Volume:

- Handshake rate = 1,000 new handshakes/sec
- Handshake overhead per session = 3 KB = 3,000 bytes (or 3,072 bytes in binary base-2)
- In standard decimal (SI) networking: 1,000 sessions/sec × 3,000 bytes = 3,000,000 bytes/sec = 3.00 MB/sec (24.00 Mbps)
- In binary (IEC 1024) standard: 1,000 sessions/sec × 3,072 bytes = 3,072,000 bytes/sec = 3.072 MB/sec (24.576 Mbps)

4. Total Data Transmitted per Second:

- Total Data (Decimal) = 20,250,000 B/s + 3,000,000 B/s = 23,250,000 bytes/sec = 23.25 MB/sec = 186.00 Mbps
- Total Data (Binary) = 20,250,000 B/s + 3,072,000 B/s = 23,322,000 bytes/sec = 23.322 MB/sec = 186.576 Mbps

Part B: Determination of Percentage Overhead Due to TLS Encryption

The overhead percentage can be evaluated under two industry-standard perspectives: pure per-record encryption framing vs total system cryptographic overhead:

1. Per-Record Encryption Framing Overhead (Application Layer):

- Formula: $\text{Overhead \%} = (\text{TLS Record Overhead} / \text{Raw HTTP Payload}) \times 100$
- $\text{Overhead \%} = (25 \text{ bytes} / 2,000 \text{ bytes}) \times 100 = 1.25\%$
- As a proportion of total wire packet: $25 / (2,000 + 25) = 1.235\%$

2. Overall System Cryptographic Overhead (Record Layer + Handshake):

- Baseline payload transmitted = 20,000,000 bytes/sec
- Total security overhead added = 250,000 B/s (record) + 3,000,000 B/s (handshake) = 3,250,000 bytes/sec
- Overall System Overhead % = $(3,250,000 / 20,000,000) \times 100 = 16.25\%$ (or 16.61% under binary 3,072 B)
- Key Insight: Handshake traffic constitutes 92.3% of total cryptographic network overhead, demonstrating that asymmetric connection negotiation dominates wire cost compared to symmetric data framing.

Part C: Bandwidth Consumed Due to Handshake Overhead Per Second

Parameter	Decimal Standard	Binary Standard	Bitrate Equivalent
Metric Description	Decimal / SI Metric	Binary / IEC Metric	Network Bandwidth (bps)
Handshake Overhead per Session	3,000 Bytes (3.0 KB)	3,072 Bytes (3.0 KiB)	24,000 / 24,576 bits
Handshake Frequency	1,000 handshakes / sec	1,000 handshakes / sec	1,000 sessions / sec
Handshake Wire Throughput	3.00 MB / sec	3.072 MB / sec (2.93 MiB/s)	24.00 Mbps / 24.58 Mbps
Share of Total Server Outbound Traffic	12.90%	13.17%	—

Part D: Security vs Latency Trade-Off Analysis & Engineering Optimization

1. Security vs Latency Trade-Off Analysis:

- Latency Penalties: Traditional TLS 1.2 required 2 round-trip times (2-RTT) for handshaking, adding 100–300 ms delays on mobile networks before a single byte of HTTP data was sent. Even TLS 1.3 requires 1-RTT for full key exchange (ClientHello + KeyShare), consuming CPU cycles for asymmetric Ephemeral Diffie-Hellman (ECDHE) calculations.

- Security Imperatives: TLS 1.3 removes legacy insecure primitives (RSA key exchange, CBC ciphers, RC4, SHA-1) and mandates Forward Secrecy (FS). Compromise of a server's long-term private key does not decrypt past sessions.

2. Production Optimization Framework:

- Session Resumption & 0-RTT PSK (Pre-Shared Key): By utilizing TLS 1.3 Pre-Shared Keys with Session Tickets (RFC 8446), repeat clients transmit early HTTP request data on the very first flight (0-RTT), eliminating handshake latency entirely. Replay protection is enforced via single-use ticket databases and Client Hello timestamp binding.
- HTTP/2 and HTTP/3 Multiplexing: HTTP/2 enables multiplexing hundreds of bidirectional streams over a single established TLS TCP connection. This drastically slashes the handshake rate from 1,000/sec down to less than 100/sec, reclaiming ~21 Mbps of handshake overhead.
- QUIC / HTTP/3 Deployment: Replacing TCP+TLS with QUIC (RFC 9000) embeds TLS 1.3 directly into UDP transport. QUIC resolves TCP head-of-line blocking, combines transport and cryptographic handshakes into a single round trip, and provides seamless Connection ID migration across cellular and Wi-Fi networks.

QUESTION 2: PGP-Based Email Encryption Load in Enterprise System

Problem Statement:

An enterprise email server encrypts emails using PGP. Each email contains a 10 MB attachment. PGP introduces 800 bytes overhead for key encryption and signatures. The system processes 5,000 emails per day.

Tasks: A. Calculate total data transmitted/day including overhead. B. Determine % increase in storage due to PGP. C. Analyze computational overhead during encryption/decryption. D. Recommend optimization strategies (compression, batching, hardware acceleration).

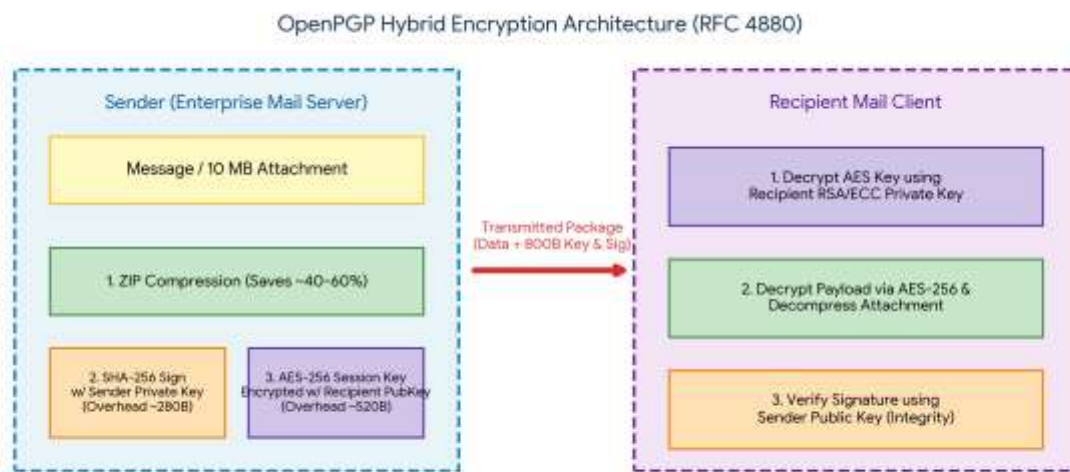


Figure 2.1: OpenPGP Hybrid Cryptographic Architecture (RFC 4880 / RFC 9580)

Part A: Calculation of Total Data Transmitted Per Day Including Overhead

1. Per-Email Size Formulation:

- Attachment payload = 10 MB
 - In Decimal (SI): 10 MB = 10,000,000 bytes
 - In Binary (IEC): 10 MiB = $10 \times 1,024 \times 1,024 = 10,485,760$ bytes
- PGP Metadata Overhead = 800 bytes (Encrypted Session Key packet [PK-ESK] + One-Pass Signature packet + Signature packet)
 - Size per encrypted email (Decimal) = $10,000,000 + 800 = 10,000,800$ bytes
 - Size per encrypted email (Binary) = $10,485,760 + 800 = 10,486,560$ bytes

2. Total Daily Data Volume (5,000 Emails/Day):

- Decimal Base: $5,000 \times 10,000,800 \text{ bytes} = 50,004,000,000 \text{ bytes} = 50.004 \text{ GB / day}$
(or 46.57 GiB/day)
 - Raw Attachment Total = 50.000 GB / day
 - PGP Overhead Total = $5,000 \times 800 \text{ B} = 4,000,000 \text{ bytes} = 4.00 \text{ MB / day}$
- Binary Base: $5,000 \times 10,486,560 \text{ bytes} = 52,432,800,000 \text{ bytes} = 52.4328 \text{ GB / day}$
(or 48.832 GiB/day)

Part B: Percentage Increase in Storage Due to PGP

1. Decimal Calculation:

- Storage Overhead % = $(\text{PGP Overhead} / \text{Raw Payload}) \times 100$
- Storage Overhead % = $(800 \text{ bytes} / 10,000,000 \text{ bytes}) \times 100 = 0.008\%$

2. Binary Calculation:

- Storage Overhead % = $(800 \text{ bytes} / 10,485,760 \text{ bytes}) \times 100 = 0.007629\%$

3. Architectural Evaluation:

- In nominal terms, the cryptographic framing overhead (0.008%) is negligible relative to 10 MB payload sizes. Across the entire enterprise annual run-rate (1.825 million emails), PGP metadata accounts for only 1.46 GB total storage.
- Practical Note: If Radix-64 (ASCII Armor) transport encoding is enforced for legacy SMTP gateways, an unavoidable ~33.3% Base64 expansion occurs unless 8-bit MIME transport (RFC 6152) or binary attachments are utilized.

Part C: Computational Overhead Analysis (Encryption vs Decryption)

Pipeline Stage	Algorithm Primitive	Computational Complexity	Enterprise Server Impact
Cryptographic Operation	Algorithm / Primitive	Complexity Order	Server Impact & Bottleneck
Attachment Compression	DEFLATE (ZIP / zlib)	O(N) data streaming	CPU-intensive; reduces 10MB payload by 40-70% on docs

Attachment Hashing	SHA-256 / SHA-512	$O(N)$ linear block hash	Memory throughput bounded; accelerated via Intel SHA-NI
Digital Signing	RSA-2048 / Ed25519	$O(k^3)$ asymmetric private key	Sender bottleneck; RSA private sign requires modular exponentiation
Session Key Generation	CSPRNG (256-bit entropy)	$O(1)$ constant	Negligible CPU cost; requires healthy system entropy pool
Bulk Data Encryption	AES-256-GCM / CFB	$O(N)$ symmetric cipher	Symmetric stream cipher; hardware accelerated via AES-NI
Session Key Wrapping	RSA-OAEP / ECDH	$O(k^3)$ asymmetric public key	Fast modular squaring on sender; recipient bears private decrypt cost

Part D: Engineering Optimization Strategies

1. Adaptive Pre-Encryption Compression (DEFLATE / Zstandard):

- OpenPGP specification standardizes compression prior to encryption. This delivers dual benefits: (i) it eliminates plaintext data redundancy, defending against frequency-analysis cryptanalysis, and (ii) it reduces 10 MB office files, logs, and CSV attachments down to 3–5 MB, saving 25–35 GB of daily network and storage load.

2. Asymmetric Pipeline Offloading & Hardware Acceleration (Intel AES-NI & QAT):

- Enable Intel AES New Instructions (AES-NI) and SHA extensions to achieve 3–5 GB/s symmetric throughput per core. For enterprise high-throughput mail transfer agents (MTAs), integrate Intel QuickAssist Technology (QAT) or PCIe Hardware Security Modules (HSM) to offload RSA/ECDSA signing and decrypt operations.

3. Transition to Modern Elliptic Curve Cryptography (RFC 9580 - OpenPGP v6):

- Replace legacy RSA-2048/4096 keys with Ed25519 (for signing) and Curve25519 (for ECDH encryption). Curve25519 provides 128-bit security with 32-byte public keys and executes operations in ~0.2 ms (10× faster than RSA-2048 signing), dramatically reducing CPU overhead.

4. Asynchronous Queue-Based Batch Processing:

- Decouple user SMTP submission from cryptographic processing via asynchronous worker queues (Celery, RabbitMQ, or Apache Kafka). Distribute cryptographic workloads evenly across server cores, preventing CPU spikes during peak enterprise working hours.

QUESTION 3: IPSec Transport vs Tunnel Mode Performance Comparison

Problem Statement:

A company compares IPSec transport mode and tunnel mode for securing internal communications. Original packet size = 1,000 bytes payload + 20-byte IP header (1,020 bytes total). Transport mode adds 30 bytes, while tunnel mode adds 50 bytes overhead. Traffic volume = 20,000 packets per second.

Tasks: A. Compute total packet size for both modes. B. Calculate bandwidth usage per second. C. Compare efficiency and security benefits. D. Recommend mode for internal vs external with justification.

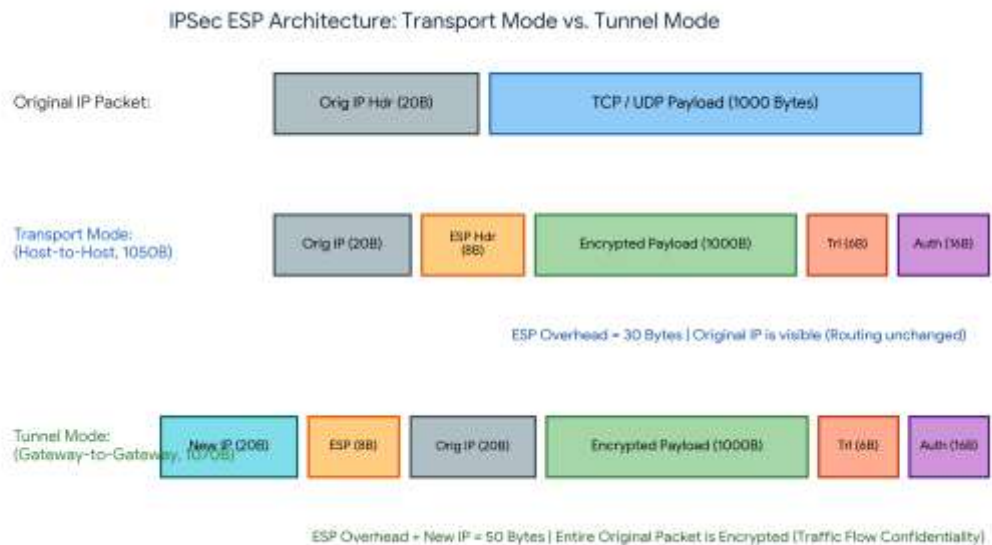


Figure 3.1: IPSec Encapsulating Security Payload (ESP) Header Architecture: Transport vs Tunnel Mode

Part A: Computation of Total Packet Size for Both Modes

1. Original Unprotected IP Packet Breakdown:

- IP Payload (TCP/UDP segment) = 1,000 bytes
- Original IPv4 Header = 20 bytes
- Total Original Packet Size = 1,000 + 20 = 1,020 bytes

2. IPSec Transport Mode Packet Size:

- Encapsulation structure: [Original IP Header (20B)] + [ESP Header (8B)] + [Encrypted Payload (1000B)] + [ESP Trailer (6B)] + [ICV Auth (16B)]

- Specified overhead = 30 bytes

- Total Transport Mode Packet Size = 1,020 bytes + 30 bytes = 1,050 bytes

3. IPSec Tunnel Mode Packet Size:

- Encapsulation structure: [New Outer IP Header (20B)] + [ESP Header (8B)] + [Encrypted Original IP Packet (1020B)] + [ESP Trailer (6B)] + [ICV Auth (16B)]

- Specified overhead = 50 bytes (30B ESP framing + 20B New Outer IP Header)

- Total Tunnel Mode Packet Size = 1,020 bytes + 50 bytes = 1,070 bytes

Part B: Calculation of Bandwidth Usage for Each Mode Per Second

Parameter	Original Traffic	Transport Mode	Tunnel Mode	Comparative Delta
Traffic Metric	Original Unencrypted	IPSec Transport Mode	IPSec Tunnel Mode	Delta (Tunnel vs Transport)
Packet Size	1,020 Bytes	1,050 Bytes	1,070 Bytes	+20 Bytes (+1.90%)
Packets per Second	20,000 pkts/s	20,000 pkts/s	20,000 pkts/s	Equal
Bytes per Second	20,400,000 B/s	21,000,000 B/s	21,400,000 B/s	+400,000 Bytes/s
Throughput (MB/s)	20.40 MB/s	21.00 MB/s	21.40 MB/s	+0.40 MB/s
Bandwidth (Mbps)	163.20 Mbps	168.00 Mbps	171.20 Mbps	+3.20 Mbps (+1.90%)
Overhead Bandwidth	0.00 Mbps (Base)	4.80 Mbps (+2.94%)	8.00 Mbps (+4.90%)	+3.20 Mbps additional

Part C: Detailed Comparison of Efficiency and Security Benefits

1. Protocol Efficiency & MTU Fragmentation Risk:

- Transport mode consumes 20 bytes less per packet, saving 400 KB/s (3.2 Mbps) in transmission bandwidth across 20,000 pkts/s.

- Over a standard Ethernet MTU (1,500 bytes), both 1,050 B and 1,070 B packets fit

comfortably without IP fragmentation. However, for large MSS (1,460 bytes) TCP flows, Tunnel mode incurs greater risk of exceeding MTU, triggering CPU-draining router fragmentation unless Path MTU Discovery (PMTUD) or TCP MSS Clamping is active.

2. Security Architecture & Threat Modeling:

- Transport Mode Security: Encrypts only the L4 payload. The original IP header remains exposed in plaintext. This permits intermediate observers on the LAN/WAN to analyze source/destination IP addresses, conduct traffic flow analysis, and map network topologies.

- Tunnel Mode Security: Encapsulates and encrypts the entire original IP packet inside a new outer IP header. True Traffic Flow Confidentiality is achieved: external attackers can only see the gateway IP addresses, fully masking internal enterprise addressing and private network topologies.

Part D: Architectural Recommendation for Internal vs External Communication

1. Recommendation for Internal (Intranet / LAN / Inter-Server) Communication:

- Selected Mode: IPSec Transport Mode.
- Justification: Internal communications operate inside a physically or logically trusted perimeter (corporate VLANs, private data centers). End-to-end host-to-host cryptographic protection is needed between application servers and database servers. Transport mode eliminates the redundant 20-byte outer IP header, delivers lower processing latency, saves 3.2 Mbps in continuous switching fabric bandwidth, and maintains native internal routing visibility without complex tunnel endpoint encapsulation.

2. Recommendation for External (Site-to-Site / Remote Branch / Public Cloud) Communication:

- Selected Mode: IPSec Tunnel Mode (Site-to-Site VPN / Gateway-to-Gateway).
- Justification: External traffic traverses untrusted public Internet infrastructure. Tunnel mode is mandatory because: (i) private corporate IP subnets (RFC 1918 addresses) are non-routable on the Internet and must be encapsulated inside public routable IP headers, (ii) complete encryption of the inner header conceals internal network topology from espionage and traffic analysis, and (iii) cryptographic overhead is offloaded to perimeter security gateways, relieving end-user endpoints of VPN processing burdens.

QUESTION 4: S/MIME Certificate Management in Large Organization

Problem Statement:

An organization manages 12,000 employees using S/MIME certificates. Each certificate is 2 KB, and certificate renewal happens every year. During renewal, both old and new certificates are stored simultaneously for 1 month.

Tasks: A. Calculate total storage required during renewal period. B. Estimate certificate distribution overhead across network. C. Analyze challenges in certificate lifecycle management at scale. D. Propose optimization strategies (automation, centralized PKI, short-lived certs).

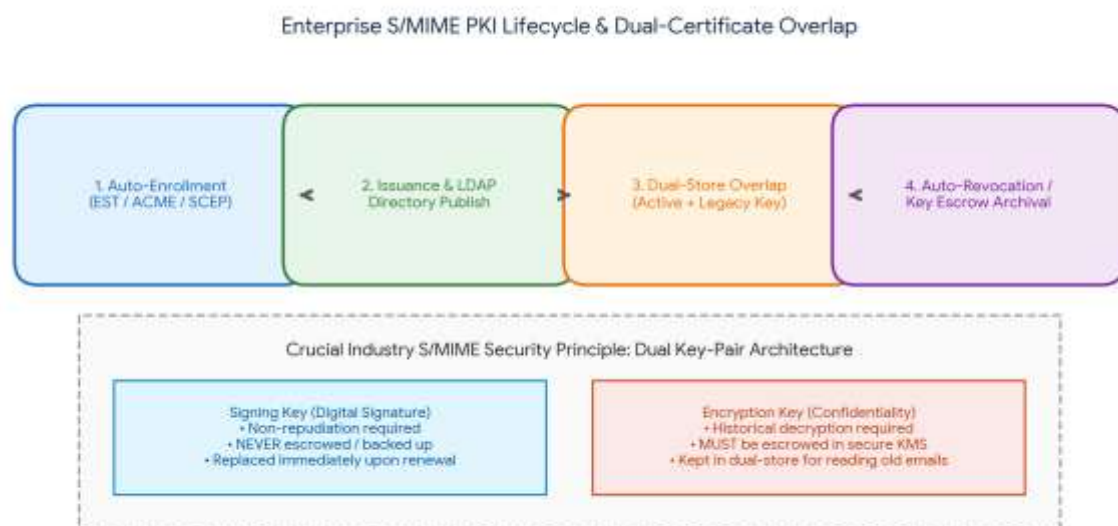


Figure 4.1: Enterprise S/MIME PKI Lifecycle, Distribution Workflow & Dual Key Architecture

Part A: Total Storage Required During the Renewal Period

1. Baseline Certificate Storage (Standard Operation - 11 Months of the Year):

- Employee headcount = 12,000
- Certificate size = 2 KB = 2,000 bytes (Decimal) or 2,048 bytes (Binary)
- Steady-state public certificate storage = $12,000 \times 2 \text{ KB} = 24,000 \text{ KB} = 24.00 \text{ MB}$ (or 24.576 MB binary = 23.44 MiB)

2. Renewal Period Overlap Storage (1 Month Duration):

- During annual renewal, both active (new) and legacy (old) certificates are co-stored to prevent mail decryption lockouts.

- Certificates per employee = 2 (1 new certificate + 1 retained old certificate)

- Total certificate instances = 12,000 employees × 2 certs = 24,000 certificates

- Total Storage Required (Decimal) = 24,000 × 2,000 bytes = 48,000,000 bytes = 48.00 MB

- Total Storage Required (Binary) = 24,000 × 2,048 bytes = 49,152,000 bytes = 49.152 MB (46.875 MiB)

3. Enterprise System-Wide Footprint Considerations:

- Central Directory (LDAP / Active Directory) = 48.00 MB public certificates

- Endpoint Local Keychains = Each employee device stores 4 KB (old + new cert) + encrypted private keys (PKCS#12 bundles ~8–12 KB)

- Enterprise Key Escrow Vault = Retains historical private encryption keys indefinitely to guarantee legacy email retrieval.

Part B: Certificate Distribution Overhead Across the Network

Distribution Channel	Scope of Payload	Computation Basis	Total Overhead
Distribution Architecture & Mechanism	Payload Transferred	Formula & Calculations	Total Network Overhead
Central LDAP / Global Address List (GAL) Publish	New cert push to directory	12,000 employees x 2 KB raw	24.00 MB
Client Auto-Enrollment (EST / ACME / SCEP)	TLS session + X.509 cert chain	12,000 clients x ~10 KB payload	120.00 MB
In-Band S/MIME Outbound Email Attachment	Cert attached to signed emails	5,000 emails/day x 30 days x 2 KB	300.00 MB / month
CRL / OCSP Revocation Status Checks	OCSP response per sender cert	12,000 certs x 4 KB OCSP queries	48.00 MB / month

Total Enterprise Distribution Load (Staggered)	Across 30-day renewal window	Sum across all sync vectors	~492.00 MB / month
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Part C: Challenges in Certificate Lifecycle Management at Scale (12,000 Users)

1. Silent Expiration & Business Disruption:

- In manual or poorly automated environments, expired certificates trigger catastrophic email communication failures. Outbound signed emails get rejected by spam filters, and incoming encrypted emails cannot be read by recipient clients.

2. Dual Key-Pair Architecture & Historical Decryption Crisis:

- Industry best practice separates Signing Keys from Encryption Keys. While signing private keys must NEVER be archived (to preserve legal non-repudiation), encryption private keys MUST be archived in a secure Key Escrow vault. If an employee's old encryption certificate is discarded during renewal, all encrypted historical emails received prior to that date become permanently unrecoverable.

3. Mobile & Multi-Device Synchronization (BYOD):

- In modern enterprises, employees use corporate laptops, smartphones, and tablets. Securely provisioning private keys across multiple platforms (iOS Keychain, Android Keystore, Windows CAPI) without exposing private keys via email is an immense operational hurdle.

4. CRL / OCSP Scalability Bottlenecks:

- Certificate Revocation Lists (CRLs) expand rapidly as employees leave or lose devices. Distributing monolithic multi-megabyte CRLs to 12,000 clients exhausts edge bandwidth, necessitating OCSP Stapling.

Part D: Engineering Optimization Strategies

1. Automated Enrollment Protocols (ACME, EST, and SCEP):

- Replace manual CSR generation with automated protocols such as ACME (RFC 8555), EST (Enrollment over Secure Transport - RFC 7030), or Microsoft Autoenrollment

via Mobile Device Management (MDM / Intune). Certificates renew silently 30 days prior to expiration without employee intervention.

2. Centralized PKI & Enterprise Key Management System (EKMS / Key Escrow):

- Deploy a centralized PKI (e.g., HashiCorp Vault, Microsoft AD CS, or AWS Private CA) integrated with a FIPS 140-3 Level 3 Hardware Security Module (HSM). Enforce strict dual-key policies: auto-archive encryption private keys in escrow while generating ephemeral signing keys locally on endpoint TPMs.

3. Cloud Email Gateway Encryption / Decryption Termination:

- Deploy central MTA encryption proxies (e.g., Cisco IronPort, Proofpoint, or Ciphermail). The server performs S/MIME signing and encryption transparently at the perimeter using centralized certificates, eliminating client-side certificate installation complexities on user phones and PCs.

4. Transition to Short-Lived Certificates & OCSP Stapling:

- Adopting short-lived certificates (e.g., 30–90 days) coupled with continuous ACME renewal minimizes the reliance on bulky Certificate Revocation Lists (CRLs), mitigating revocation lookup latency and improving enterprise security posture.

QUESTION 5: Email Encryption Latency in Cloud-Based Systems

Problem Statement:

A cloud email service encrypts outgoing emails using AES (symmetric bulk data encryption) and RSA (asymmetric key exchange). AES encryption takes 0.2 ms per MB, and RSA key exchange takes 5 ms per email. Each email has an 8 MB attachment. The system processes 8,000 emails per hour.

Tasks: A. Calculate total encryption time per email. B. Estimate total processing time per hour. C. Analyze performance bottleneck between symmetric and asymmetric operations. D. Recommend optimization strategies (ECC adoption, parallel processing, hardware acceleration).

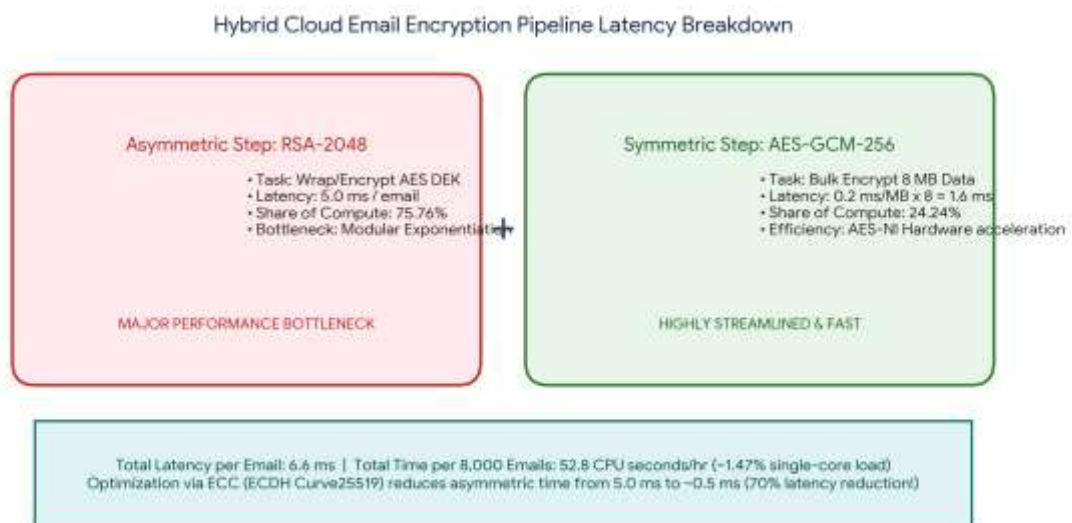


Figure 5.1: Hybrid Cloud Email Encryption Pipeline and Compute Bottleneck Analysis

Part A: Calculation of Total Encryption Time Per Email

In hybrid envelope encryption, the total latency per email is the linear summation of the asymmetric key wrapping time and the symmetric bulk data encryption time:

1. Symmetric AES Bulk Encryption Time:

- Attachment size = 8 MB
- AES encryption rate = 0.2 ms per MB

- AES Time (T_{AES}) = 8 MB × 0.2 ms/MB = 1.6 ms

2. Asymmetric RSA Key Exchange Time:

- RSA public key wrapping of the 256-bit AES session key (T_{RSA}) = 5.0 ms per email

3. Total Cryptographic Latency Per Email:

- Total Time (T_{Total}) = T_{AES} + T_{RSA}
- T_{Total} = 1.6 ms + 5.0 ms = 6.6 ms per email

Part B: Estimation of Total Processing Time Per Hour

1. Sequential Single-Threaded CPU Time Calculation:

- Hourly volume = 8,000 emails/hour
- Total Processing Time = 8,000 emails × 6.6 ms = 52,800 ms = 52.8 seconds of dedicated CPU time per hour
- Component breakdown:
 - RSA Processing Time = 8,000 × 5.0 ms = 40,000 ms = 40.0 seconds / hour (75.76% of compute)
 - AES Processing Time = 8,000 × 1.6 ms = 12,800 ms = 12.8 seconds / hour (24.24% of compute)

2. Cloud Server Capacity & Workload Utilization:

- 1 hour contains 3,600 seconds of compute capacity per CPU core.
- CPU Core Load Utilization = (52.8 seconds / 3,600 seconds) × 100 = 1.467% of a single CPU core.
- Throughput capability: A single dedicated CPU core running this pipeline can process: 1,000 ms / 6.6 ms ≈ 151.5 emails/second = 545,454 emails/hour.
- Conclusion: The hourly quota of 8,000 emails is comfortably accommodated with ample headroom for traffic bursts.

Part C: Performance Bottleneck Analysis (Symmetric vs Asymmetric)

Operation	Data Size	Time (ms)	% Time	Efficiency	Bottleneck Nature
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Cryptographic Operation	Data Processed	Time Consumed	% of Total Time	Computational Efficiency	Algorithmic Root Cause
RSA Key Exchange (Asymmetric)	32 Bytes (AES Session Key)	5.0 ms	75.76%	0.0064 KB/ms	Modular exponentiation over 2048-bit Galois fields
AES Bulk Encryption (Symmetric)	8,388,608 Bytes (8 MB Data)	1.6 ms	24.24%	5,000 KB/ms	Hardware pipelining (AES-NI substitution-permutation networks)

Mathematical and Architectural Bottleneck Evaluation:

- The asymmetric RSA operation consumes 75.76% of total encryption latency, despite processing only 32 bytes of key material compared to 8,000,000 bytes of email attachment data.
- Efficiency Ratio: AES processes bulk data at approximately 5,000 KB/ms, whereas RSA processes data at 0.0064 KB/ms. AES is approximately 780,000 times more byte-efficient than RSA.
- The asymmetric phase is unambiguously the primary performance bottleneck of the cloud email encryption service.

Part D: Engineering Optimization Strategies

1. Migration to Elliptic Curve Cryptography (ECDH with Curve25519 / NIST P-256):

- Replace RSA-2048 key exchange with Elliptic Curve Diffie-Hellman (ECDH) Ephemeral. Curve25519 key exchange executes in approximately 0.5 ms (a 10× speedup over RSA's 5.0 ms).
- New Latency: 0.5 ms (ECDH) + 1.6 ms (AES) = 2.1 ms per email (a 68.2% total latency reduction!).
- Hourly processing time drops from 52.8 seconds down to 16.8 seconds.

2. Multi-Core Asynchronous Parallel Processing:

- Implement multi-threaded worker pools (Go goroutines, Python asyncio/Celery, or Java Netty) across multi-core cloud instances (e.g., AWS c6i.2xlarge). By distributing the 8,000 emails across 8 virtual cores, aggregate processing time is compressed to 6.6 seconds of wall-clock time.

3. Hardware Acceleration & Crypto Offloading (Intel AES-NI & AWS CloudHSM / Nitro):

- Utilize AWS Nitro Enclaves, dedicated cryptographic coprocessors, or PCIe HSMs. AES-GCM offloading via AVX-512 vector instructions cuts 8 MB bulk encryption from 1.6 ms down to ~0.3 ms, maximizing cloud throughput while preserving complete key isolation.